



A traditional Ethernet-enabled system (a) is built around an external PHY [physical layer of the open systems interconnection (OSI) model] requiring numerous passives and other components. When the Ethernet PHY is integrated into the MCU (b), board space requirements drop substantially, as do power consumption and system cost.

multiple low-power operating modes and a 32kHz oscillator that consumes less than one micro-ampere."

Texas Instruments' (TI) CC430 family is another example of a wireless MCU that provides tight integration between the MCU core, peripherals, software and a sub-GHz radio frequency transceiver. TI's CC2541 is another power-optimised wireless

MCU example, which enables BLE and proprietary 2.4GHz network nodes to be built with low total bill of material costs.

MCUs with growing interfaces

MCUs today have a lot of interfaces that enable hooking up various kinds of sensors and actuators. The inter-

faces include general-purpose input/output (GPIO), analogue in, analogue out, pulse width modulation (PWM), universal serial bus (USB) host and device, inter-integrated circuit (I²C), serial peripheral interface (SPI), universal asynchronous receiver/transmitter (UART), RS232, built-in analogue-to-digital converter (ADC) and built-in digital-to-analogue con-

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